Co-Optimizing Sales of Energy and Capacity in a Hydropower Scheduling Model

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***Abstract*—This paper describes a model for optimal scheduling of hydroelectric systems for a price-taking producer selling energy and capacity to separate markets. The model is based on a combination of stochastic dynamic programming (SDP) and stochastic dual dynamic programming (SDDP), and treats inflow to reservoirs and energy prices as stochastic variables. It allows sales of capacity at a deterministic sequence of capacity reserve prices. Thus, the sales of energy and capacity is co-optimized within the SDDP framework.**

The presented model is tested on a Norwegian watercourse where the producer sells energy to the day-ahead market and capacity to the primary reserve markets. When adding the possibility to sell capacity in the model, the results show that less water is used during winter and more during summer/autumn in order to sell capacity.

***Index Terms*—Hydroelectric power generation, Power generation economics, Optimization.**

I. Introduction

The future European electricity system will be more integrated across country borders, and include a larger share of renewable intermittent generation than what is the case today. This development will e.g. be driven by a stronger transmission grid, environmental targets set by the European Union and decisions on downscaling of nuclear generation capacity.

The flexibility of hydropower allows for efficient balancing of intermittent production. By fully utilizing this flexibility, hydropower producers can optimize the use and allocation of available capacity in the different markets. Thus, the value of flexible hydropower generation is likely to increase by participating in multiple markets. The importance of the different types of market products may change significantly from what hydropower producers in the Nordic market are familiar with. Today, the producers primarily benefit from selling power in the day-ahead (spot) market. However, the inherit flexibility of hydropower enables active contribution in balancing markets as well, which in turn will challenge the way hydropower systems traditionally are operated and scheduled. The choice of utilizing the capacity for production or as reserve will e.g. impact the strategic valuation and scheduling of resources. That is, the consideration of multiple markets will impact the way the *water values* are calculated, and existing scheduling tools and methods need to take this into account.

Numerous solution strategies has been applied to the hydropower scheduling problem, see e.g. [1] for a thorough review. Stochastic dynamic programming (SDP) has proved to be well suited for systems with relatively few reservoirs, but will suffer from high computation times when considering realistic multi-reservoir systems. In order to avoid the curse of dimensionality limiting the use of the SDP algorithm, an approach known as stochastic dual dynamic programming (SDDP) was presented in [2]. Unlike the case with SDP, there is no need to fully discretize the state space with the SDDP algorithm. SDDP is a sampling-based variant of multi-stage Benders decomposition, where an outer approximation of a convex future expected cost function is constructed iteratively for each time-stage by adding Benders cuts. Thus, the overall optimization problem can be decomposed into small linear programming (LP) problems that can be solved independently. Currently, SDDP seems to be the state-of-the-art method for solving the long- and medium-term scheduling problem in regions where hydropower is the dominant technology for producing electric power, see e.g. [3] and [4]. Later research related to the SDDP algorithm has e.g. focused on convergence properties [5], strategic bidding [6], risk aversion [7], emission constraints [8] and integration of pumped-storage and wind power [9].

In a liberalized market, hydropower producers will typically act as price-takers trying to optimize the utilization of hydro reservoirs to maximize their expected profit. For this purpose, [10] and [11] introduced a hybrid SDP/SDDP method. The basic idea is to combine the strengths of the SDP and the SDDP methods to retain the convexity of the problem while dealing with the curse of dimensionality. In this hybrid method, the spot price forecasts are treated as discrete Markov Chains through SDP. Other state variables, such as reservoir levels and inflows, are continuously approximated in the SDDP scheme.

In this paper a method for incorporating the possibility to sell capacity in the hybrid SDP/SDDP algorithm is proposed and evaluated. Emphasis is put on the Norwegian primary reserve market which seems to offer a certain potential for ad-

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ditional revenue for hydropower producers, particularly in the low-load (summer) season. The proposed method is applied to a case study where the optimal schedule for a watercourse is found considering participation in both the day-ahead and primary reserve markets.

II. The Primary Reserve Market

In the Nordic system, the transmission system operators (TSOs) are responsible for matching supply and demand of electricity in real time. In order to ensure this balance the TSOs need to be able to acquire balancing services, both in terms of capacity and energy. By balancing services we refer to primary, secondary and tertiary reserves. The Norwegian TSO (Statnett) acquires balancing services of all three types through market-based approaches using the marginal pricing principle [12]. In the following we will focus on the market for primary reserves.

Momentary imbalances between production and demand will firstly be regulated by use of primary regulation reserves, which may be traded in the primary reserve market. The system frequency is controlled by automatic activation of frequency-controlled reserves for normal operation (FCR-N). FCR-N contributes to both upward and downward regulation in the frequency band 49.90-50.10 Hz. Such reserves are currently assured by the droop setting in the turbine governors for generators exceeding 10 MVA. That is, generators that do not participate in the primary reserve market will still participate in the primary regulation. In previous years, Statnett would ask Norwegian generators to increase the droop [13] in case of insufficient reserves. In that sense, primary control was considered a free service and there was no market for this type of service. In 2008 two primary reserve markets were established; a weekly and daily market. The weekly market concerns capacity procurement for 6 time blocks, whereas the daily market concerns hourly procurement. These markets are operated by Statnett according to conditions stated in [14].

Fig. 1 shows the average daily procured capacity in both of these markets in 2013. There is a strong seasonal pattern in the weekly market; Relatively large amounts of capacity are procured in the low-load season, between weeks 20 and 30. This pattern indicates that in low-load periods with low power prices, there are clear incentives for the TSO to ensure sufficient amounts of spinning reserve at an early phase. Thus, volumes are moved from the daily to the weekly market to ensure that the system requirement is met in the summer season.

Fig. 2 shows both the FCR-N price in the daily market and the spot price for the Norwegian price area NO2 in 2013. The weekly FCR-N price follows the same pattern as the daily price, but is slightly higher at summer time. In the summer season there seems to be a potential for additional revenue through the FCR-N markets for producers in this area. Producers targeting this potential are more likely to keep their units spinning during summer time.

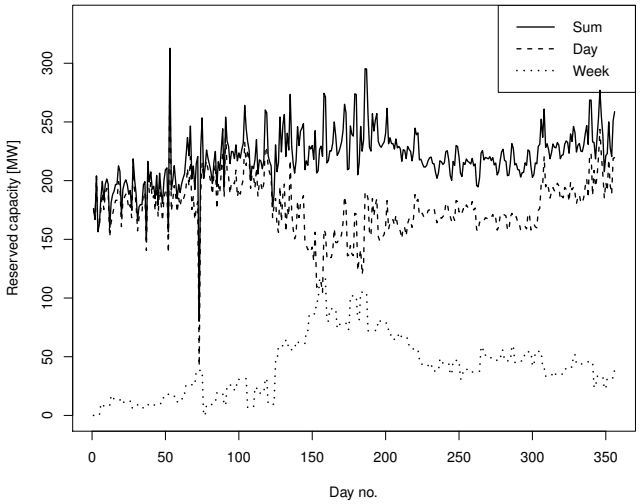


Fig. 1. Sum of procured capacity in the weekly and daily FCR-N markets in 2013. Data points are average values per day. Capacities in the two markets are shown as dotted lines and the sum as a solid-drawn line.

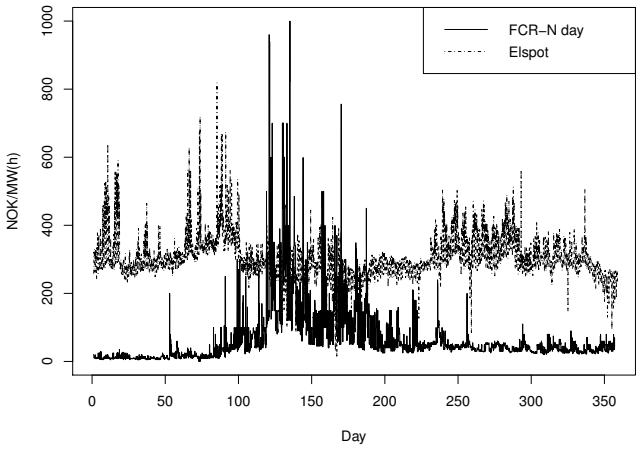


Fig. 2. Prices from the FCR-N daily market and the spot market for price area NO2 in 2013.

III. Model Description

The overall optimization problem seeks to maximize the profit of a hydropower producer selling both energy to the spot market and reserves to the primary reserve market (denoted FCR market in the following). The problem is solved by a combination of SDDP and SDP, using the algorithm described in [11]. The sum of weekly inflow to reservoirs and the weekly average spot price are considered stochastic variables. A planning period of one year and a stochastic time-resolution of one week is applied. That is, at the beginning of each week all stochastic variables are known for that week. It is assumed that the producer is a price-taker in both the spot and FCR

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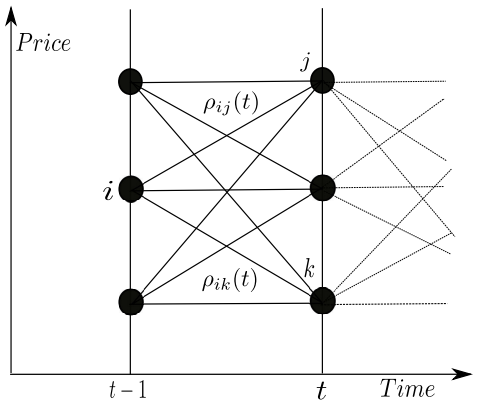


Fig. 3. Stochastic model for weekly average spot price.

markets.

A. Price Modelling

Both the price processes for the spot and FCR markets are represented in the model. The spot market price is considered as an exogenous stochastic variable, and price forecasts can typically be obtained from a long-term market model. Normally the weekly average spot price will show a strong serial correlation; thus, it is necessary to include a price state in the model's system state description. As discussed in [11], the price state violates the convexity requirement of the SDDP algorithm. This section briefly presents the price model used in SDDP applications, where the price process is modelled as a Markov chain using discrete states (price points), and embedded in the SDDP algorithm as in ordinary dynamic programming.

The spot price model is illustrated in Fig. 3, indicating transitions from price point i in stage $t-1$ to price points j and k in stage t. The transition probabilities $\rho_{ij}(t)$ and $\rho_{ik}(t)$ are computed prior to the optimization, as elaborated in [15]. Cuts from the SDDP algorithm are stored for each price point in each time stage, and cannot be shared among time stages. When making a cut for stage $t-1$ and price point i, the decomposed optimization problem for stage t is solved for all N_P price points in that stage, and the contributions are averaged over the transition probabilities ρ_{ip}, where $p = 1, ..., N_P$.

Assume that a set of scenarios are available, describing both the spot and FCR prices. Both price processes could then in principle be described in a discrete Markov chain, but now in a two-dimensional state space. The transition matrix could be identified in a similar manner as for the spot price model, retaining possible correlations between the two price processes. However, due to the fact that the Nordic/Norwegian market design for balancing markets has been subject to frequent improvements lately and there are few years of complete data, this work will use recent observations of FCR prices from the Norwegian primary reserve market as deterministic input.

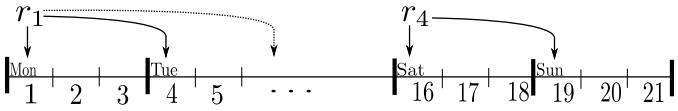


Fig. 4. Intra-week time resolution. Energy is sold per time step and capacity per block.

B. Time Resolution

Each week is divided into 21 time steps and capacity can be sold in 3 blocks covering weekdays (night, day and evening), and 3 blocks covering weekends (night, day and evening). This segmentation of time steps is in line with the current market design for the Norwegian weekly primary reserve market [14].

The intra-week time resolution is illustrated in Fig. 4. At the beginning of each week, all prices are known for that week. The spot price point, representing the weekly average spot price level, is used to scale an average spot price profile, giving one unique spot price for each of the 21 time steps. FCR prices are provided for each of the 6 reservation blocks. Decisions regarding power production to the spot market are done for each time step, whereas decisions on sales of capacity are done per reserve block. The capacity sold in block 1 covering the night period in weekdays should therefore be a joint decision for all time steps belonging to block 1, as illustrated with r_1 in Fig. 4.

C. Decomposed Problem Structure

In this section the decomposed problem structure for co-optimzing sales of energy and capacity is described. Details on how this decomposed problem fits in to the combined SDDP/SDP algorithm can be found in [11].

The decomposed weekly decision problem is formulated as an LP problem described by (1)-(7). For a given week t we know the realization of stochastic variables for that week (inflows and spot prices). The amount of energy sold to the spot market and capacity to the FCR market is co-optimized for the whole water course, and we assume that there are no demand obligations.

$$\max \left\{ \alpha_t + \sum_{k \in \mathcal{K}} \lambda_k^E \sum_{h \in \mathcal{H}} p_{hk} + \sum_{b \in \mathcal{B}} \lambda_b^R \sum_{h \in \mathcal{H}} \tau_b r_{hb} \right\} \tag{1}$$

$$v_{hk} + \sum_{s \in \mathcal{S}(h)} q_{hks} + q_{hk}^s - \sum_{j \in \Omega_h} \left(\sum_{s \in \mathcal{S}(j)} q_{jks} + q_{jk}^s \right) = v_{h,k-1} + I_{hk} \quad , h \in \mathcal{H}, k \in \mathcal{K} \tag{2}$$

$$p_{hk} = \sum_{s \in \mathcal{S}(h)} \eta_{hs} q_{hks} \quad , h \in \mathcal{H}, k \in \mathcal{K} \tag{3}$$

$$r_{hb(k)} - \frac{1}{\tau_k} p_{hk} \leq 0 \quad , h \in \mathcal{H}, k \in \mathcal{K} \tag{4}$$

$$r_{hb(k)} + \frac{1}{\tau_k} p_{hk} \leq P_h^{max} \quad , h \in \mathcal{H}, k \in \mathcal{K} \tag{5}$$

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$$\alpha_t \leq \sum_{h \in \mathcal{H}} \pi_{h\ell} v_{hk} + c_\ell \quad , k = K \quad , \ell \in \mathcal{L}_i \tag{6}$$

$$0 \leq r_{hb} \leq \frac{2\Delta f P_h^{max}}{\delta_h} \quad , h \in \mathcal{H}, b \in \mathcal{B} \tag{7}$$

Variables:

α_t	Future expected revenue function, in €;
p_{hk}	Generated electricity for station h, in MWh;
r_{hb}	Reserved capacity for station h, in MW;
v_{hk}	Reservoir level for reservoir h, in Mm^3;
q_{hks}	Discharge through station h, segment s, in Mm^3;
q^s_{hk}	Spillage from reservoir h, in Mm^3.

Parameters:

λ^E_k	Spot price in time step k, in €/MWh;
I_{hk}	Inflow to reservoir h, in Mm^3;
τ_b	Total duration of reservation block b, in hours;
τ_k	Duration of time step k, in hours;
λ^R_b	FCR price in block b, in €/MW;
η_{hs}	Energy equivalent for station h, in MWh/Mm^3;
P^{max}_h	Maximum capacity in station h, in MW;
π^ℓ_h	Cut coefficient for reservoir h and cut ℓ, in €/Mm^3;
c^ℓ	Cut right-hand side for cut ℓ, in €;
Δf	Frequency band for FCR, 0.1 Hz;
δ_h	Droop setting for station h, in percentage;
K	Last time step in week;
Ω_h	Set of reservoirs upstream h;
$\mathcal{H}$	Set of hydropower reservoirs/stations;
$\mathcal{S}(h)$	Set of discharge segments for station h;
$\mathcal{K}$	Set of time steps in week;
$\mathcal{B}$	Set of reservation blocks in week;
$\mathcal{L}_i$	Set of cuts for a given price point i.

The objective (1) is to maximize the income from both markets, subject to constraints (2)-(7). The water balance equation for a specific reservoir h in a water course is formulated in (2). For each plant, the energy p_{hk} is produced in a time step k according to (3). Sold capacity r_{hb} in block b should be spinning and symmetric. The spinning requirement is taken care of in (4), ensuring that the producer do not offer more reserve capacity than what is already spinning in a station. Eqn. (5) ensures that the generation capacity sold in the two markets is within the station's installed capacity limit. The revenues obtained for the current decision period (week) are balanced against the future expected revenue function, which is constrained by cuts in (6). These cuts are built and accumulated in each backward iteration of the SDDP algorithm, and are treated by relaxation when solving the LP problem.

All variables have non-negativity constraints and most of them have upper bounds. The reservation variable r_{hb} is of special interest in this study, and is constrained as shown in (7). The capacity sold to the FCR market cannot exceed the physical limit dictated by the droop setting. Thus, the model takes a pre-defined droop setting δ_h for each station as input, which in turn limits the station's market access according to (7).

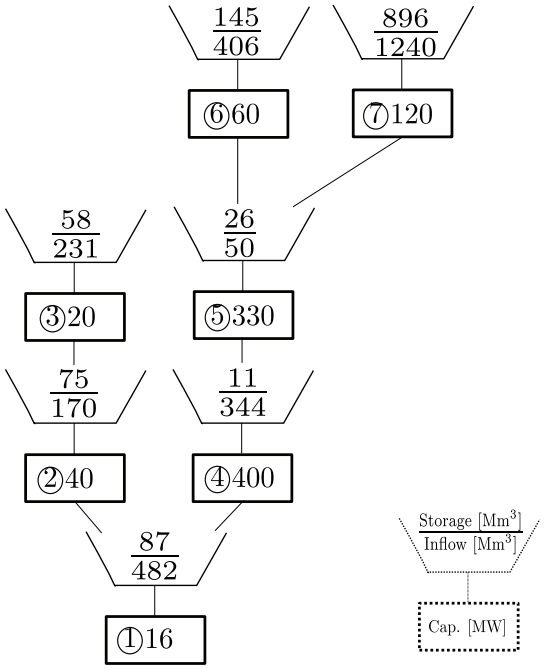


Fig. 5. Watercourse topology and technical data.

Note that the model presentation is somewhat simplified compared to the actual implementation, omitting e.g. penalty variables related to the water balance (2) needed to ensure that the stochastic model has complete recourse. Furthermore, inflows are sampled from a autoregressive model both in the forward and backward iteration of the SDDP algorithm, so that convergence is guaranteed.

IV. Case Study

The proposed SDDP/SDP scheduling model was tested on a Norwegian watercourse comprising 7 hydropower reservoirs with corresponding power stations, with a total capacity of 986 MW. An illustration of the topology and technical characteristics is provided in Fig. 5. For each reservoir shown in the figure the average annual inflow and storage capacity are stated, both in Mm^3. Each power station is identified with a number and its installed capacity in MW. The water course is located within the Norwegian price area NO2, and it is assumed that the operator will act as a price-taker when selling energy to the spot market and spinning capacity to the primary reserve market. A scheduling horizon of one year was applied with weekly stochastic time resolution and 21 time steps within each week. A set of spot price scenarios was obtained from the EMPS model, which is a fundamental hydrothermal market model [16]. The discrete spot price model was calculated, as discussed in Sec. III-A. FCR prices are from the Norwegian weekly market (prices for the daily market are similar, and shown in Fig. 2).

Two cases were defined; one where the producer was allowed to sell in both markets (referred to as the ***dual-market*** case), and one where only sales to the energy market was

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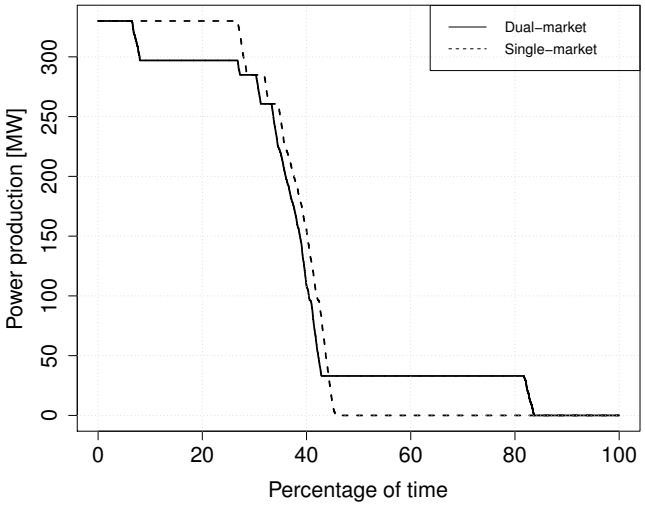


Fig. 6. Hydropower generation from station no. 5.

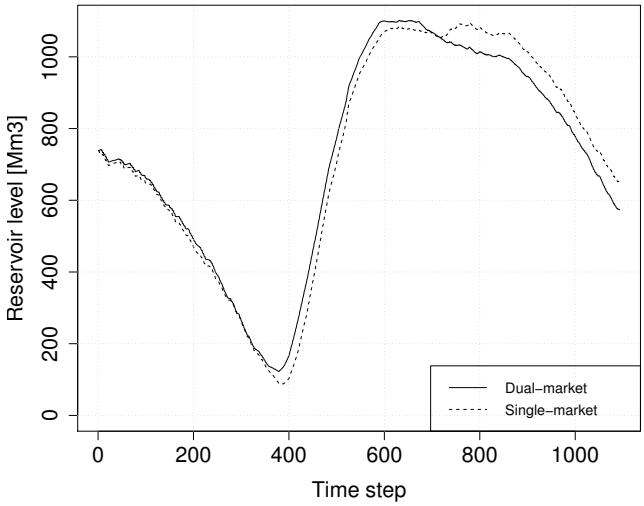


Fig. 7. Average reservoir trajectories for the sum of all reservoirs.

allowed (referred to as the ***single-market*** case). A set of end-value cuts was used to account for the value of stored water at the end of the planning horizon. This set was similar in both cases.

Model runs were carried out with 5 price points representing the stochastic spot price, a total of 80 forward samples and 8 backward samples in the SDDP algorithm and a maximum of 100 iterations. The model was implemented in `C++`, using the dual simplex algorithm from the Gurobi library [17]. All tests were carried on an Intel Core i7-Q740 processor with 1.73 GHz and 16 GB RAM. A single run required approximately 42 hours, and the model demonstrated good convergence properties.

A. System Operation

Comparing the two cases, one observes significant differences in the operation of hydropower stations. This is illustrated in Fig. 6, where the duration curves for production from station no. 5 for both cases are shown. Station 5 has its maximum efficiency at 79% (260 MW) and can, according to (7), offer 33 MW to the FCR market. Fig. 6 shows that the station is kept spinning at 33 MW for a substantial amount of time in order to sell capacity to the FCR market. Conversely, in time periods where the single-market case show generation at maximum capacity, 33 MW is frequently withheld from spot and sold as FCR in the dual-market case.

The change in hydropower generation can also been seen when looking at the reservoir trajectories. Fig. 7 shows the average trajectories for the sum of all reservoirs for both cases. At the beginning of the year the reservoirs are emptied rapidly due to high spot prices and low inflow. However, since capacity is withheld for the FCR market, the curve is kept higher in the dual-market case. Conversely, more water is used during summer time in the dual-market case.

The dual value of (2) is often referred to as the water value for the given reservoir. For a given system state one should expect the water value to be higher in the dual-market case since a higher revenue can be expected. Fig. 8 shows the difference in average water values between the dual- and single market cases for all time steps for the 4 largest reservoirs in the system. A similar pattern is revealed for these reservoirs; the dual-market water value is lower during before spring flood (around time step 400) and higher during summer and autumn. This result can be related to the reservoir trajectories in Fig. 7. The higher water value in early spring in the single-market case is due to more aggressive use of water, i.e. running the stations at maximum output and following a lower reservoir trajectory. At summer and autumn, more water is used in the dual-market case to keep the units spinning, resulting in higher water values in this period.

B. Discussion

The revenue in the dual-market case was on average 2.58% higher than in the single-market case, which is too optimistic compared to the actual opportunities that producers see in today's market. The model overestimates the ability to trade in the primary reserve market for several reasons. First of all, it is assumed that the market always has room for the primary reserves being offered provided a pre-defined droop setting at each station. This assumption can be debated since the TSO will only procure a limited total amount of reserves. Secondly, since the presented optimization model is linear, we do not consider binary decisions regarding start/stop of power stations. The LP model requires a piecewise linear and concave relationship between power output and water discharge through the stations. Thus, stations are allowed to generate electricity at low rates at an artificially high efficiency for the purpose of delivering reserves. Thirdly, the

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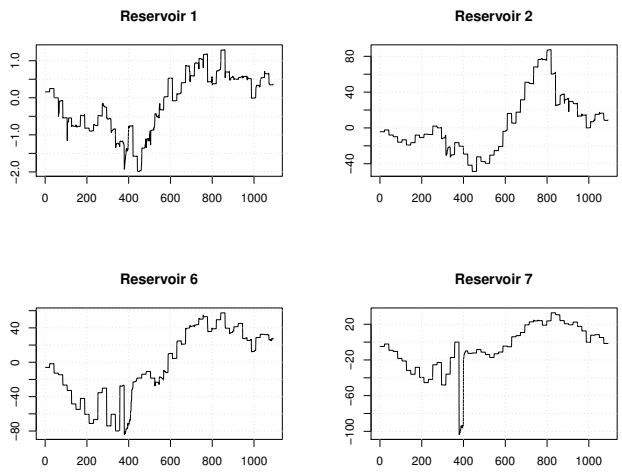


Fig. 8. Difference in average water value for the 4 largest reservoirs. Dual-market minus single-market. Water values are in €/Mm^3.

implemented decision sequence seems to give the model too much freedom to operate in the two different markets. At the beginning of the week all prices in both markets are seen by the model, allowing it to deterministically optimize the use of resources for that week. Thus, the decisions regarding the two markets are co-optimized as opposed to the sequential decision making under uncertainty that takes place in reality.

The simulated water values are important results from this analysis. In order to draw solid conclusions on the differences in water values between the two cases, the period of analysis should be extended to limit the impact of the end-value cuts. Moreover, the end-value cuts should be set independently for each case to reflect the differences in future expected profit.

V. Conclusions

A hybrid SDP/SDDP model was presented, suitable for solving the medium- and long-term hydro-thermal scheduling problem co-optimizing sales of energy and capacity. Traditionally such models only consider the energy market. This work demonstrates that a capacity market can be introduced by a modest extension of a previously presented scheduling model aimed at the energy market. Moreover, the added computational burden and model complexity is low.

Although the model extension is based on several simplifications and overestimates the profit possible to obtain, we believe it can provide useful additional insight for producers operating in multiple markets.

References

[1] J. W. Labadie, "Optimal operation of multireservoir systems: State-of-the-art review," *Journal of Water Resources Planning and Management*, vol. 130, no. 2, pp. 93–111, 2004.

[2] M. V. F. Pereira and L. M. V. G. Pinto, "Multi-stage stochastic optimization applied to energy planning," *Mathematical Programming*, vol. 52, pp. 359–375, 1991.

[3] M. E. P. Maceira, V. S. Duarte, D. D. J. Penna, L. A. M. Moraes, and A. C. G. Melo, "Ten years of application of stochastic dual dynamic programming in official and agent studies in Brazil - description of the Newave program," in *in Proc. 16th Power System Computation Conference*, Glasgow, 2008.

[4] A. Gjelsvik, B. Mo, and A. Haugstad, *Handbook of Power Systems I*. Springer, 2010, ch. Long- and medium-term operations planning and stochastic modelling in hydro-dominated power systems based on stochastic dual dynamic programming, pp. 33–55.

[5] A. Philpott and Z. Guan, "On the convergence of stochastic dual dynamic programming and related methods," *Operations Research Letters*, vol. 36, no. 4, pp. 450–455, 2008.

[6] B. Flach, L. Barroso, and M. Pereira, "Long-term optimal allocation of hydro generation for a price-maker company in a competitive market: latest developments and a stochastic dual dynamic programming approach," *IET Generation, Transmission & Distribution*, vol. 4, no. 2, pp. 299–314, 2010.

[7] A. Philpott and V. de Matos, "Dynamic sampling algorithms for multi-stage stochastic programs with risk aversion," *European Journal of Operational Research*, vol. 218, no. 2, pp. 470–483, 2012.

[8] S. Rebennack, B. Flach, M. V. F. Pereira, and P. M. Pardalos, "Stochastic hydro-thermal scheduling under CO_2 emissions constraints," *IEEE Transactions on Power Systems*, vol. 27, no. 1, pp. 58–68, 2012.

[9] A. Helseth, A. Gjeslvik, B. Mo, and U. Linnet, "A model for optimal scheduling of hydro thermal systems including pumped-storage and wind power," *IET Generation, Transmission & Distribution*, vol. 7, no. 12, pp. 1426 – 1434, 2013.

[10] A. Gjelsvik and S. W. Wallace, "Methods for stochastic medium-term scheduling in hydro-dominated power systems," SINTEF Energy Research, Tech. Rep. TR A4438, 1996.

[11] A. Gjelsvik, M. M. Belsnes, and A. Haugstad, "An algorithm for stochastic medium-term hydrothermal scheduling under spot price uncertainty," in *Proc. 13th Power System Computation Conference*, Trondheim, Norway, 1999.

[12] http://www.statnett.no/en/Market-and-operations/.

[13] T. Kristiansen, "The Nordic approach to market-based provision of ancillary services," *Energy Policy*, vol. 35, pp. 3681–3700, 2007.

[14] Statnett, "Vilkår for tilbud, aksept, rapportering og avregning i marked for primærreserver," www.statnett.no, 2013.

[15] B. Mo, A. Gjelsvik, A. Grundt, and K. Kåresen, "Optimisation of hydropower operation in a liberalised market with focus on price modelling," in *In proc. of IEEE Power Tech Conference*, Porto, Portugal, 2001.

[16] O. Wolfgang, A. Haugstad, B. Mo, A. Gjelsvik, I. Wangensteen, and G. Doorman, "Hydro reservoir handling in norway before and after deregulation," *Energy*, vol. 34, no. 10, pp. 1642–1651, 2009.

[17] I. Gurobi Optimization, "Gurobi optimizer reference manual," 2015. [Online]. Available: http://www.gurobi.com

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